

MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Fundamental of Algebra

(English Medium)

EXERCISE (O-1)

Straight Objective Type

1. $\sqrt{6 + \sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}}} =$

- (A) 3 (B) 2 (C) 1 (D) ± 3

FM0008

2. If $x = 8 - \sqrt{60}$, then $\frac{1}{2} \left[\sqrt{x} + \frac{2}{\sqrt{x}} \right] =$

- (A) $\sqrt{5}$ (B) $\sqrt{3}$ (C) $2\sqrt{5}$ (D) $2\sqrt{3}$

FM0009

3. If $\frac{4+3\sqrt{5}}{4-3\sqrt{5}} = a+b\sqrt{5}$, a, b are rational numbers, then $(a, b) =$

- (A) $\left(\frac{61}{29}, \frac{-24}{29}\right)$ (B) $\left(\frac{-61}{29}, \frac{24}{29}\right)$ (C) $\left(\frac{61}{29}, \frac{24}{29}\right)$ (D) $\left(\frac{-61}{29}, \frac{-24}{29}\right)$

FM0010

4. The square root $5 + 2\sqrt{6}$ is :

- (A) $\sqrt{3} + 2$ (B) $\sqrt{3} - \sqrt{2}$ (C) $\sqrt{2} - \sqrt{3}$ (D) $\sqrt{3} + \sqrt{2}$

FM0011

5. If $\frac{4}{2+\sqrt{3}+\sqrt{7}} = \sqrt{a} + \sqrt{b} - \sqrt{c}$, then which of the following can be true -

- (A) $a = 1, b = 4/3, c = 7/3$ (B) $a = 1, b = 2/3, c = 7/9$
(C) $a = 2/3, b = 1, c = 7/3$ (D) $a = 7/9, b = 4/3, c = 1$

FM0012

6. The numerical value of $\left(x^{1/a-b}\right)^{1/a-c} \times \left(x^{1/b-c}\right)^{1/b-a} \times \left(x^{1/c-a}\right)^{1/c-b}$ is (a, b, c are distinct real numbers)

- (A) 1 (B) 8 (C) 0 (D) None

FM0013

7. $(1^3 + 2^3 + 3^3 + 4^3)^{-3/2} =$

- (A) 10^{-3} (B) 10^{-2} (C) 10^{-4} (D) 10^{-1}

FM0014

8. $\left(7^{\left(-\frac{1}{2}\right)} \times 5^2\right)^2 \div \sqrt{25^3} =$

- (A) $\frac{5}{7}$ (B) $\frac{7}{5}$ (C) 35 (D) $-\frac{5}{7}$

FM0015

9. $(2d^2e^{-1})^3 \times \left(\frac{d^3}{e}\right)^{-2} =$

- (A) $8e^{-2}$ (B) $8e^{-3}$ (C) $8e^{-1}$ (D) $8e^{-4}$

FM0016

10. If $x^y = y^x$ and $x = 2y$, then the values of x and y are ($x, y > 0$)

- (A) $x = 4, y = 2$ (B) $x = 3, y = 2$ (C) $x = 1, y = 1$ (D) None of these

FM0017

11. If $(a^m)^n = a^{m^n}$, then express 'm' in the terms of n is ($a > 0, a \neq 1, m > 1, n > 1$)

- (A) $n^{\left(\frac{1}{n-1}\right)}$ (B) $n^{\left(\frac{1}{n+1}\right)}$ (C) $n^{\left(\frac{1}{n}\right)}$ (D) None

FM0018

12. If $a = x + \frac{1}{x}$, then $x^3 + x^{-3} =$

- (A) $a^3 + 3a$ (B) $a^3 - 3a$ (C) $a^3 + 3$ (D) $a^3 - 3$

FM0019

13. If $\left(\sqrt[3]{4}\right)^{2x+\frac{1}{2}} = \frac{1}{32}$, then $x =$

- (A) -2 (B) 4 (C) -6 (D) -4

FM0020

14. If $(5 + 2\sqrt{6})^{x^2-3} + (5 - 2\sqrt{6})^{x^2-3} = 10$, then all possible values of x are

- (A) $-2, 2$ (B) $\sqrt{2}, -\sqrt{2}$ (C) $2, +\sqrt{2}$ (D) $2, -2, \sqrt{2}, -\sqrt{2}$

FM0021

15. If $3^{2x^2} - 2 \cdot 3^{x^2+x+6} + 3^{2(x+6)} = 0$ then the value of x is

- (A) -2 (B) 3 (C) Both (A) and (B) (D) None of these

FM0022

16. How many integers in between 100 to 1500 (both inclusive) are multiples of 5 or 11 ?

- (A) 408 (B) 26 (C) 382 (D) 380

FM0023

17. Square root of $4 + \sqrt{15}$ is equal to

- (A) $\frac{\sqrt{3} + \sqrt{5}}{2}$ (B) $\sqrt{\frac{3}{2}} + \sqrt{\frac{5}{2}}$ (C) $\sqrt{\frac{5}{2}} - \sqrt{\frac{3}{2}}$ (D) None of these

FM0024

18. If $A = \{(x, y) \mid xy = 8 \text{ and } x, y \in \mathbb{Z}\}$, then $n(A) =$

- (A) 4 (B) 8 (C) 12 (D) 16

FM0025

19. The set of all real numbers x that satisfy $\frac{x-2}{x+2} > \frac{2x-3}{4x-1}$

(A) $(-\infty, -2) \cup \left(\frac{1}{4}, \frac{3}{2}\right) \cup (2, \infty)$

(B) $\left(-2, \frac{1}{4}\right) \cup \left(\frac{3}{2}, 2\right)$

(C) $(-\infty, -2) \cup \left(\frac{1}{4}, 1\right) \cup (4, \infty)$

(D) $\left(-2, \frac{1}{4}\right) \cup (1, 4)$

FM0026

20. The set of all real numbers x that satisfy $\frac{x^2-4}{x^2-5x+6} \leq 0$

(A) $[-2, 3]$

(B) $[-2, 3)$

(C) $(-\infty, -2] \cup (3, \infty)$

(D) None of these

FM0027

21. If $\frac{(x+3)^2(x-1)^9(x+1)^5}{(x-3)(x-5)^4(x-6)^5} \leq 0$, then number of possible integral values of x is -

(A) 6

(B) 3

(C) 4

(D) 5

FM0028

22. If $(\sqrt{2}+1)^x + (\sqrt{2}-1)^x - 2\sqrt{2} = 0$, then sum of all possible values of x is

(A) 0

(B) 1

(C) 2

(D) 3

FM0029

23. If $x^2 + y^2 + 4z^2 - 6x - 2y - 4z + 11 = 0$, then xyz is equal to

(A) $3/2$

(B) 4

(C) 6

(D) 3

FM0030

24. If $\frac{\sqrt{3}+4\sqrt{2}}{4\sqrt{2}-\sqrt{3}} = \frac{a+b\sqrt{6}}{c}$, then the value of $a+b+c$ (where $a, b, c \in \mathbb{N}$ and are relatively prime)

(A) 70

(B) 72

(C) 50

(D) 40

FM0034

25. If $x^2 + 4y^2 + z^2 - 2xy - 2yz - zx = 0$ then $x : y : z$ equals

(A) $1 : 2 : 1$

(B) $2 : 1 : 2$

(C) $1 : 2 : 3$

(D) $1 : 1 : 2$

FM0035

EXERCISE (O-2)

More than one correct

1. If $x = \sqrt{7\sqrt{7\sqrt{7\sqrt{7}\dots}}}$ where $x, y > 0$

$y = \sqrt{20 + \sqrt{20 + \sqrt{20 + \dots}}}$ then which of the following is/are correct.

(A) $x + y = 12$ (B) $x - y = 3$ (C) $x^2 + y^2 = 74$ (D) $x^2 - y^2 = 24$

FM0037
2. If $8x^3 + 27x^2 - 9x - 50$ is divided by $(x + 2)$ then remainder is λ then

(A) $\frac{3\lambda + 4}{10}$ is equal to 4

(B) If $\lambda = \frac{p}{q}$ then $(p + q)$ is divisible by 13 (where p & q are coprime)

(C) λ is a natural number

(D) $(\lambda - 2)$ is divisible by 3

FM0039
3. If $f(x) = ax^3 + bx^2 + cx + d$, where $a, b, c, d \in \mathbb{R}, a \neq 0$ then which of the following option(s) is(are) correct ?

(A) if $a + b + c + d = 0$ then $x - 1$ is factor of $f(x)$.

(B) if $a + b = c + d$ then $x + 1$ is factor of $f(x)$.

(C) if $a + c = b + d$ then $x + 1$ is factor of $f(x)$.

(D) none of these

FM0040
4. If $x = a$ and $x = b$ are the two roots of the equation $9^x - 4 \times 3^{x+1} + 27 = 0$ then

(A) $a + b = 3$ (B) $(a - b)^2 = 1$ (C) $\frac{a}{b} + \frac{b}{a} = \frac{5}{2}$ (D) $a + b = 4$

FM0041
5. The real values of ' x ' satisfying the equation $(4 + \sqrt{15})^{x^2-x-1} + (4 - \sqrt{15})^{x^2-x-1} = 8$ is/are :

(A) 0 (B) 1 (C) -1 (D) -2

FM0042
6. The value of $(4 + 1)(4^2 + 1)(4^4 + 1)(4^8 + 1) + \frac{1}{3} = \frac{2^\lambda}{3}$ where $\lambda \in \mathbb{N}$ then the factor(s) of ' λ ' is/are

(A) 4 (B) 8 (C) 16 (D) 32

FM0075
7. If $x = \frac{6mn}{m+n}$, then value of $\left(\frac{x+3m}{x-3m} + \frac{x+3n}{x-3n}\right)$, (where $m \neq n \neq 0$) is less than or equal to

(A) 1 (B) 2 (C) 3 (D) 4

8. If $x = \sqrt{6 + \sqrt{6 + \sqrt{6 + \dots \infty}}}$ and $y = \sqrt{6\sqrt{6}\sqrt{6}\dots\infty}$, then

- (A) $x = 2y$ (B) $x + y = 9$ (C) $x^2 - y = 3$ (D) x is a prime number

FM0129

9. Value of $\left(\frac{x^{\frac{2}{3}}y^{-\frac{1}{2}}}{yx^{-\frac{2}{3}}} \div \sqrt{\frac{xy^{-2}}{yx^{-1}}} \right)^6$ is not equal to

- (A) x^2 (B) x (C) x^3 (D) x^4

FM0130

10. If $\frac{1}{2-\sqrt{3}} - \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{5}{3-\sqrt{2}} = a + b\sqrt{2}$; where $a, b \in \mathbb{Q}$ then

(Note : $\mathbb{Q}$ denotes set of rational numbers)

- (A) $a + b = \frac{41}{7}$ (B) $a + b = \frac{39}{7}$ (C) $\frac{a}{b} = 5$ (D) $\frac{a}{b} = \frac{29}{12}$

FM0131

11. If $\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{3^2}\right)\left(1 + \frac{1}{3^4}\right) \dots \left(1 + \frac{1}{3^{512}}\right) = \frac{1}{18} \left(\frac{3^a - 1}{3^{1021}}\right)$, then

- (A) sum of the digits of number 'a' is 5. (B) sum of the digits of number 'a' is 7.
(C) last digit of number 'a' is 3. (D) 'a' is divisible by 8.

FM0132

12. If $x(y + z) = 16$; $y(z + x) = 66$; $z(x + y) = 70$ and $x, y, z > 0$, then $(x + y + z)$ is less than

- (A) 16 (B) 17 (C) 26 (D) 27

FM0133

13. If $x^2 - 1$ is a factor of $8x^3 + qx^2 + px + 2$, then $(p + 4q)$ is less than

- (A) -8 (B) -2 (C) -10 (D) -16

FM0134

14. Which of the following option(s) is(are) true

- (A) sum of all roots of the equation $|9 - x| = 5$ is 4
(B) sum of all roots of the equation $|9 - x| = 5$ is 18
(C) sum of all roots of the equation $|x - 9| = 5$ is 18
(D) sum of all roots of the equation $|x - 9| = 5$ is 14

FM0135

15. If $f(x)$ is monic polynomial of degree 3 such that $f(0) = 0$, $f(1) = 1$, $f(2) = 2$ then which of the following option(s) is/are true ?

- (A) $f(4) = 4$ (B) $|f(-1)| = 1$ (C) $f(-1) = -7$ (D) $f(-2) = -26$

FM0136

EXERCISE (O-3)

Paragraph for question no. 1 to 3

$$\text{If } (5 + 2\sqrt{6})^{x^2-8} + (5 - 2\sqrt{6})^{x^2-8} = 10, x \in \mathbb{R}$$

On the basis of above information, answer the following questions :

1. Number of solution(s) of the given equation is/are-
 (A) 1 (B) 2 (C) 4 (D) infinite FM0043
2. Sum of positive solutions is
 (A) 3 (B) $3 + \sqrt{7}$ (C) $2 + \sqrt{5}$ (D) 2 FM0043
3. If $x \in (-3, 5]$, then number of possible values of x , is-
 (A) 1 (B) 2 (C) 3 (D) 4 FM0043

Paragraph for question no. 4 to 6

A polynomial $p(x)$ when divided by $(x - 1)$, $(x + 1)$ and $(x + 2)$ gives remainder 5, 7 and 2 respectively. If $p(x)$ is divided by $(x^2 - 1)$ and $(x - 1)(x + 2)$ gives remainder as another polynomial $R(x)$ and $r(x)$ respectively. Then

4. The value of $R(50)$ is :
 (A) 34 (B) -44 (C) 44 (D) 104 FM0044
5. The value of $r(100)$ is :
 (A) 34 (B) 44 (C) 54 (D) 104 FM0044
6. The value of $r(10)$ is :
 (A) 4 (B) 14 (C) 24 (D) 35 FM0137

Matching list type

7. Match the list

List-I

- (P) The units digit of $2^7 \times 3^{10}$ is
 (Q) The number of prime factors of $6^{100} \times 15^{200}$ is
 (R) Number of solutions of $xy = 35$ in natural numbers with $x > y > 1$ is
 (S) Remainder when $x^{100} - 3x^{99} + 2x^{98} + 3x - 2$ is divided by $x - 2$ is equal to

List-II

- (1) 1 FM0045
 (2) 2 FM0046
 (3) 3 FM0047
 (4) 4 FM0048

Codes :

- (A) P-1, Q-2, R-3, S-4 (B) P-2, Q-3, R-1, S-4
 (C) P-2, Q-3, R-2, S-3 (D) P-1, Q-4, R-1, S-2

8. Match the list

List-I

List-II

(P) If $x = \sqrt{5} - 2$ then value of $2x^3 + 11x^2 + 10x + 4$ is equal to (1) 8

FM0049

(Q) If $x = \sqrt{5} + 2$ then value of $2x^3 - 7x^2 - 6x + 7$ (2) 7

FM0050

(R) If $\left(\left(\sqrt{11+2\sqrt{30}}\right) + \left(\sqrt{10+4\sqrt{6}}\right)\right)^2 = p + \sqrt{q} + \sqrt{r} + \sqrt{s}$ (3) 161
where p, q, r, s are integers & q, r, s are not perfect squares the p + q + r + s is equal to

FM0051

(S) If $\left(\left(\sqrt{11+2\sqrt{30}}\right) - \left(\sqrt{10+4\sqrt{6}}\right)\right)^2 = \alpha - \sqrt{\beta}$ (4) 977
where α, β are integers & β is not a perfect square then $\alpha^2 + \beta$ is equal to

FM0052

Codes :

(A) P-1, Q-2, R-3, S-4

(B) P-2, Q-3, R-1, S-4

(C) P-2, Q-1, R-3, S-4

(D) P-2, Q-1, R-4, S-3

Answer the question question 9 and 10 by appropriately matching the lists based on the information given

List-I

List-II

(I) If $2\sqrt{x+5} = x+2$, then integral value of x

(P) 3

(II) The number of real solution of the equation $x^2 + 5|x| + 4 = 0$, is

(Q) -4

(III) The real solution of the equation $(x-2)^2 + |x-2| - 2 = 0$, is

(R) -1

(IV) The number of real solution(s) $|x+2| = 2(3-x)$, is

(S) 0

(T) 4

(U) 1

9. Which of the following is only **CORRECT** option ?(A) $I \rightarrow R, U$ (B) $II \rightarrow S, T$ (C) $I \rightarrow Q, S$ (D) $II \rightarrow S$

FM0053

10. Which of the following is only **INCORRECT** option ?(A) $III \rightarrow P, U$ (B) $IV \rightarrow U$ (C) $III \rightarrow Q, T$ (D) $III \rightarrow P$

FM0053

EXERCISE (O-4)

Numerical Grid Type

1. If $a - \frac{1}{a} = \frac{1}{2}$ then $\left(4a^2 + \frac{4}{a^2}\right)$ is equal to FM0054
2. If polynomial $Ax^3 + 4x^2 + Bx + 5$ leaves same remainder, when divided by $x - 1$ and $x + 2$ respectively then value of $3A + B$ is equal to FM0055
3. If $9^x + 6^x = 2.4^x$ then the value of $\frac{x^3 + 64}{5}$ FM0056
4. If $a + b + c = 6$ & $a^2 + b^2 + c^2 = 14$ and $a^3 + b^3 + c^3 = 36$ then the value of $3\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$ FM0057
5. If $x + y + z = 12$ & $x^2 + y^2 + z^2 = 96$ and $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 36$. Find the value of $\frac{x^3 + y^3 + z^3}{4}$ FM0104
6. If $x = \sqrt{4 - 2\sqrt{3}}$ and $y = \sqrt{9 - 4\sqrt{5}}$ then the value of $(\sqrt{5}x - \sqrt{3}y)^2$ is equal to $(a - b\sqrt{c})$ where a, b, c are coprime numbers then $a + b + c$ is equal to (where 'c' is an odd integer) FM0105
7. If $x, y, z \in \mathbb{R}$ and $121x^2 + 4y^2 + 9z^2 - 22x + 4y + 6z + 3 = 0$ then value of $x^{-1} - y^{-1} - z^{-1}$ is equal to FM0106
8. If $\frac{p}{q} = \frac{x+2}{x-2}$, then find the value of $\frac{p^2 - q^2}{p^2 + q^2} \times \frac{x^2 + 4}{x}$. ($x \neq 0$) FM0108
9. $\frac{3x + \sqrt{9x^2 - 5}}{3x - \sqrt{9x^2 - 5}} = 5$ find 'x' FM0110
10. Value of $\left(\frac{\sqrt{6}}{\sqrt{3} - \sqrt{2} + \frac{\sqrt{6}}{\sqrt{3} - \sqrt{2} + \frac{\sqrt{6}}{(\sqrt{3} - \sqrt{2}) + \dots}}} \right)^2$ is FM0111

JEE-MAINS/ JEE-ADVANCE

1. The expression $\frac{12}{3+\sqrt{5}+2\sqrt{2}}$ is equal to

(a) $1-\sqrt{5}+\sqrt{2}+\sqrt{10}$

(b) $1+\sqrt{5}+\sqrt{2}-\sqrt{10}$

(c) $1+\sqrt{5}-\sqrt{2}+\sqrt{10}$

(d) $1-\sqrt{5}-\sqrt{2}+\sqrt{10}$

FM0117

2. If $x < 0$, $y < 0$, $x + y + \frac{x}{y} = \frac{1}{2}$ and $(x + y)\frac{x}{y} = -\frac{1}{2}$, then $x = \dots$ and $y = \dots$

FM0118

3. The equation $x - \frac{2}{x-1} = 1 - \frac{2}{x-1}$ has

(A) no root

(B) one root

(C) two equal roots

(D) infinitely many roots

FM0119

4. Find all real values of x which satisfy $x^2 - 3x + 2 > 0$ and $x^2 - 2x - 4 \leq 0$.

FM0123

5. Find the set of all x for which $\frac{2x}{(2x^2 + 5x + 2)} > \frac{1}{(x+1)}$.

FM0124

6. The sum of all real roots of the equation $|x - 2|^2 + |x - 2| - 2 = 0$ is ...

FM0125

7. The number of real solutions of the equation $|x|^2 - 3|x| + 2 = 0$ is

(A) 4

(B) 1

(C) 2

(D) 3

FM0126

8. If S is the set of all real x such that $\frac{2x-1}{2x^3+3x^2+x}$ is positive, then S contains

(A) $\left(-\infty, -\frac{3}{2}\right)$

(B) $\left(-\frac{3}{2}, -\frac{1}{4}\right)$

(C) $\left(-\frac{1}{4}, \frac{1}{2}\right)$

(D) $\left(\frac{1}{2}, 3\right)$

(E) None of these

FM0127

9. Let $y = \sqrt{\frac{(x+1)(x-3)}{(x-2)}}$

Find all the real values of x for which y takes real values.

FM0128

ANSWER KEY

Do yourself-1

1. B 2. C 3. A 4. D 5. C
6. (i) 1 (ii) 0 (iii) 1 (iv) 1 (v) 8
7. 2 8. (a) $-\frac{47}{44}$, (b) $-\frac{9}{44}$ 9. 5 10. 26
11. 1 12. $x = y = z = \frac{a}{3}$ and if a is integral multiple of 6 then $x = y = z = \pm a/3$

Do yourself-2

1. (i) $(x - y)(x + y)(x^2 + y^2)$ (ii) $(3a - 2x + y)(3a + 2x - y)$
(iii) $(2x + 3y)(2x - 3y - 3)$
2. (i) $(2x - 3y)(4x^2 + 6xy + 9y^2)$ (ii) $(2x - 5y)[4x^2 + 10xy + 25y^2 + 1]$
3. (i) $(x + 8)(x - 5)$ (ii) $(x - 8)(x + 5)$ (iii) $(x + 7)(x - 2)$
(iv) $(x - 4)(x + 1)$ (v) $(x - 3)(x + 1)$ (vi) $(3x - 4)(x - 2)$
(vii) $(4x + 7)(3x - 5)$ (viii) $(3x - 2)(x - 1)$ (ix) $(x - 1)(3x - 4)$
(x) $(7x - 1)(x - 1)$ (xi) $(x - 2)(2x - 13)$ (xii) $(a - 3)(3a + 2)$
(xiii) $(2a + 1)(7a - 3)$
4. (i) $(a - b - 1)(a + b - 3)$ (ii) $(x^2 - 6x + 18)(x^2 + 6x + 18)$
(iii) $(x^2 - y + 1)(x^2 + y + 1)$ (iv) $(2a^2 - a - 1)(2a^2 + a - 1)$
(v) $(2x^2 - 6x + 9)(2x^2 + 6x + 9)$ (vi) $(x^4 - x^2 + 1)(x^4 + x^2 + 1)$
5. (i) $(x - 1)(x - 2)(x - 3)$ (ii) $(x + 1)(x + 3)(2x + 1)$
(iii) $(x - 1)(x - 2)(2x - 3)$ (iv) $(x^2 + 2)(x^2 + 1)\left(x - \sqrt{3}\right)\left(x + \sqrt{3}\right)$
(v) $3(x + y)(y + z)(z + x)$
6. (i) $(2a^2 - a + 1)(4a^4 + 2a^3 - a^2 + a + 1)$
7. (i) $(x^2 + 5x + 1)(x^2 + 5x + 9)$
(ii) $2(2x^2 + 2x + 3)(4x^2 + 4x - 9)$
(iii) $(x^2 + 5x - 22)(x + 1)(x + 4)$

8. (i) $(x - y - 5)(x - y + 4)$. (ii) $(x + 2y - 2)(x + 2y + 4)$.
 (iii) $16(3x + 4)(x + 1)$. (iv) $(x^{\frac{1}{3}} + 2)(x^{\frac{1}{3}} - 1)$.
 (v) $(x - 2)(x^2 + 2x + 4)(x + 1)(x^2 - x + 1)$.
 (vi) $(x^2 + 16)(x - 1)(x + 1)$.
 (vii) $(x + 1)(x - 2)(x - 3)(x + 2)$. (viii) $(x^2 + 5x + 16)(x + 6)(x - 1)$.
 (ix) $-2(a^2 + 1)(a^2 + 5)$. (x) $(x + y + z - 3)^2$.
 (xi) $(2x - 1)(x - 2)(x + 1)^2$. (xii) $(x^2 + 2)(x + 1)(x - 1)(x^4 + x^2 + 1)$.

Do yourself-3

1. (a) 7 (b) 47 (c) 18 4. 104 5. $-\sqrt{2}$ 6. 1
 8. (a) 0, 4 (b) no solution (c) no solution

Do yourself-4

1. 1 2. -20 3. 4 4. -2 5. $3x + 4$
 6. $\ell = 2, m = -18$ 7. $\ell = -1, m = -3$ 8. $a = 3, b = 1$
 9. $3x^3 + 4x^2 - 5x - 2$ 10. 6 11. $a = 8$ and $b = 5$ 12. 22

Do yourself-5

1. 1 or -8 2. 1, 32 3. $\pm\sqrt{2}, \pm\sqrt{\frac{2}{3}}$
 4. 3 or 4 5. $x = 1, x = \frac{3 \pm \sqrt{5}}{2}$ 6. $x = \pm 1, 2, -\frac{1}{2}$
 7. 0, -1 8. 2, 4 9. ± 1
 10. -4, 4 11. -4, 1 12. -6, 1
 13. $x = -3 \pm \sqrt{5}$ 14. $\pm 1, 1 \pm \sqrt{2}$
 15. $(A) \rightarrow (U); (B) \rightarrow (U); (C) \rightarrow (T); (D) \rightarrow (P, Q, R, S, T, U, V); (E) \rightarrow (R, T); (F) \rightarrow (S)$

Do yourself-6

1. $x = 1, y = 3$ 2. $x = 5, y = 1$
 3. $x = 5, y = 3$ 4. $x = 2, y = 1$ or $x = -1, y = -2$
 5. $x = 4, y = -2$ or $x = -2, y = 4$ 6. $x = -6, y = -2$ or $x = -4, y = -4$
 7. $x = 3, y = 2$ or $x = -3, y = -2$ 8. $x = 5, y = 1$ or $x = -5, y = -1$
 9. $x = \frac{11}{13}, y = -\frac{24}{5}$ 10. $x = 3, y = 2$ or $x = -3, y = -2$
 11. $x = 4, y = \pm 3$ 12. $x = -1, y = 3$ or $x \in \mathbb{R}, y = 2$
 13. $x = 2, y = -1$ or $x = -1, y = 2$ 14. $(x, y) \in (3, 1), (-3, -1), (1, 3), (-1, -3)$

Do yourself-7

1. (a) F (b) T (c) F (d) T (e) T (f) F
 2. (a) $\frac{17}{21}$ (b) $\frac{37}{41}$ 3. 3^{30} 4. $(2, \infty)$ 5. $3 + \sqrt{14}$

Do yourself-8

1. $4\sqrt{2}$ 2. -12 3. 4

Do yourself-9

1. $\frac{20}{27}$ 2. $21 : 35 : 65$ 3. $\frac{sp}{p+q}, \frac{sq}{p+q}$ 4. $\frac{a^4}{b^4}$

Do yourself-10

1. $(-1, 4) - \{1\}$ 2. $\left(-\frac{1}{4}, \frac{5}{6}\right)$
 3. $(-\infty, -2)(-2, -1) \cup (1, \infty)$ 4. $\left(-\infty, \frac{1}{2}\right) \cup (2, \infty) - \{4\}$
 5. $[-1, 2) - \{0\}$ 6. $(-1, 1] \cup [3, \infty) \cup \{2\}$
 7. $\left(-\frac{1}{2}, 1\right)$ 8. $(-2, -1) \cup (0, 5) - \{2\}$
 9. $[-\sqrt{2}, \sqrt{2}] \cup [3, 4) - \{-1\}$ 10. $(-2, -1) \cup (1, 2)$
 11. $\left(-\infty, -\frac{\sqrt{63}}{2}\right) \cup \left(-3, \frac{\sqrt{63}}{2}\right) \cup (4, \infty)$ 12. $\left(\frac{3}{4}, 1\right) \cup (7, \infty)$
 13. $(-\sqrt{2}, 0) \cup (1, \sqrt{2}) \cup (2, \infty)$ 14. $(-\infty, -3] \cup [2, \infty)$
 15. $(-2, -1)$ 16. $\left[-\frac{1}{4}, \frac{1}{3}\right]$
 17. (i) $[-3, 1]$ (ii) $[-7, 1]$
 (iii) $[-3, 3]$ (iv) $(-\infty, 0) \cup (3, +\infty)$
 (v) $(-\infty, 3) \cup (4, +\infty)$ (vi) $(-\infty, -2) \cup (-2, -1) \cup (1, +\infty)$
 (vii) $(-1, 5)$ (viii) $[1, 3] \cup (5, +\infty)$
 (ix) $\left(-\frac{9}{2}, -2\right) \cup (3, \infty)$ (x) $(-1, 1) \cup (4, 6)$

(xi) $(-1, 1) - \{0\}$

(xii) $\left(-\infty, -\frac{\sqrt{7}}{2}\right) \cup \left(-1, \frac{\sqrt{7}}{2}\right) \cup \left(\frac{4}{3}, \infty\right)$

(xiii) $(-\infty, -2] \cup (-1, 4)$

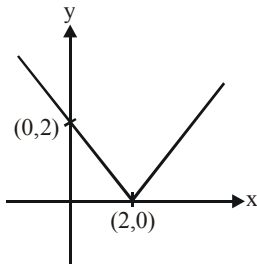
(xiv) $(-\infty, -5) \cup (1, 2) \cup (6, +\infty)$

(xv) $(-\infty, -3) \cup (-2, 3)$

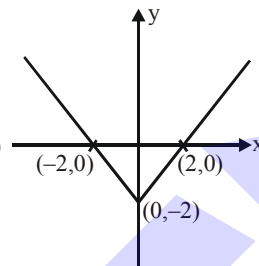
(xvi) $(-\infty, -2) \cup (-1, 3) \cup (4, \infty)$

Do yourself-11

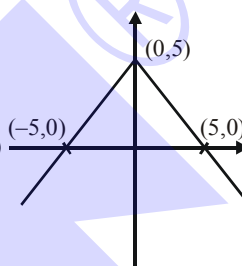
1. (i)



(ii)



(iii)



2. $\left\{-\frac{3}{2}, -\frac{7}{2}\right\}$

3. $\{-1, 6\}$

4. ϕ

5. ϕ

6. $\left\{-\frac{25}{3}, \frac{17}{3}\right\}$

7. $\{0, 3\}$

8. $\left\{-8, -\frac{2}{5}\right\}$

9. $\left\{\frac{6}{5}, 18\right\}$

10. ϕ

11. $\left\{1, \frac{7}{2}\right\}$

12. ϕ

13. $\{1, -2\}$

14. $\left\{0, -\frac{24}{7}\right\}$

15. $\{-2, 0\}$

16. $\left\{-\frac{5}{2}, \frac{3}{2}\right\}$

17. $\{-2\}$

18. $[0, \infty)$

19. $\{-3, -1\}$

20. ϕ

21. $\mathbb{R}$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	A	D	D	A	A	A	A	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	D	D	C	C	B	B	C	D
Que.	21	22	23	24	25					
Ans.	D	A	A	B	B					

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C,D	A,B,C	A,C	A,B,C	A,B,C	A,B,C,D	B,C,D	B,C,D	B,C,D	A,D
Que.	11	12	13	14	15					
Ans.	B,D	C,D	A,B,C	B,C	C,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	C	B	D	B	B	D	D	C

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	9.00	4	12.80	5.5	216.50	36	16	4	$x = 1$	2

EXERCISE # JEE-MAIN/ADVANCED

- B
- $x = -\frac{1}{4}, y = -\frac{1}{4}$
- A
- $x \in [1 - \sqrt{5}, 1) \cup (2, 1 + \sqrt{5}]$
- $(-2, -1) \cup \left(-\frac{2}{3}, -\frac{1}{2}\right)$
- 4
- A
- A, D
- $x \in [-1, 2) \cup [3, \infty)$